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PLU 3743

香港中學文憑考試

Hong Kong Diploma of Secondary Education Examination

Economics

2023 Question Papers

(with marking schemes and comments on candidates' performance)



香港考試及評核局
Hong Kong
Examinations and
Assessment Authority

Introduction

This publication provides a comprehensive report on the Economics examination in the 2023 Hong Kong Diploma of Secondary Education Examination. It covers various aspects of the examination, including the objectives, contents and format of the examination, the question papers, the marking schemes, and the chief examiners' comments on candidates' performance in the live examination. Some statistics are also included.

Through this publication, we hope to provide useful information and feedback to assist teachers in their teaching of the subject and students in their learning. Similar publications are also available for other subjects in the 2023 HKDSE. Besides, a series of publications and resource materials related to the HKDSE are also available for reference to schools, candidates and the public. These include information related to examination administration, assessment requirements, examination-related statistics and feedback provided to schools and other stakeholders. For details, please visit the Authority's website (http://www.hkeaa.edu.hk/en/HKDSE/info_corner/hkdse_publications_materials/).

We would like to take this opportunity to express our gratitude to all the examination personnel who were involved in the development of question papers and the assessment of candidates' performance. Without their contributions, it would not have been possible for us to conduct the examination smoothly.

The Examination

Assessment Objectives

The objectives of this assessment are to evaluate students' abilities and achievements in the following:

1. to know and understand fundamental economic concepts and theories;
2. to apply such concepts and theories to explain real world situations, especially the Hong Kong economy;
3. to understand and interpret economic information presented in various forms;
4. to acquire the basic tools of economic analysis;
5. to analyze economic issues;
6. to evaluate arguments, proposals and policies from different perspectives and make informed judgments; and
7. to present ideas clearly, and in a well-reasoned manner, illustrating answers with diagrams and examples.

Synopsis of curriculum contents

For details, please refer to the Economics Curriculum and Assessment Guide (Secondary 4 to 6) jointly prepared by the Curriculum Development Council and the Hong Kong Examinations and Assessment Authority.

Compulsory Part

1. Basic Economic Concepts
2. Firms and Production
3. Market and Price
4. Competition and Market Structure
5. Efficiency, Equity and the Role of Government
6. Measurement of Economic Performance
7. National Income Determination and Price Level
8. Money and Banking
9. Macroeconomic Problems and Policies
10. International Trade and Finance

Elective Part (CANCELLED)

~~Elective Part 1: Monopoly Pricing, Anti-competitive Behaviours and Competition Policy; OR~~
~~Elective Part 2: Extension of Trade Theory, Economic Growth and Development~~

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2023

ECONOMICS PAPER 1

8.30 am – 9.30 am (1 hour)

INSTRUCTIONS

1. Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.
2. When told to open this book, you should check that all the questions are there. Look for the words '**END OF PAPER**' after the last question.
3. All questions carry equal marks.
4. **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
5. You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
6. No marks will be deducted for wrong answers.

Not to be taken away before the
end of the examination session

There are 45 questions in this paper. Choose the **BEST** answer for each question.

1. Mr Cheung bought restaurant premises 2 years ago for \$5 000 000 and sells 'two-dish-rice' meal boxes at the premises. Recently, his monthly profit has decreased from \$30 000 to \$10 000 due to the opening of new restaurants nearby. A businessman offers to rent Mr Cheung's restaurant premises for a monthly rental of \$15 000.

Which of the following would be included in Mr Cheung's opportunity cost of continuing to operate his 'two-dish-rice' business at the premises?

- A. the \$5 000 000 purchase price of the restaurant premises
 - B. the \$10 000 monthly profit
 - C. the \$15 000 monthly rental
 - D. the \$20 000 drop in monthly profit
2. Jacky, a Secondary 6 student in Hong Kong, is going to travel to Korea for his graduation trip. Which of the following will decrease his cost of having the graduation trip to Korea?
- A. Hong Kong dollar appreciates against South Korean Won.
 - B. The return flight from Korea to Hong Kong is delayed.
 - C. The number of confirmed cases of COVID-19 in Korea has increased drastically.
 - D. The services of the hotel in Korea are better than his expectation.
3. A band in Hong Kong adopted a real-name registration system for the sale of its concert tickets. These tickets were not transferrable so that the chance of ticket scalping (炒賣) was reduced.

Which of the following statements about the above real-name registration system is/are correct?

- (1) It is a mechanism to tackle the problem of 'what to produce'.
- (2) The private property rights over the tickets would be weakened under the system.
- (3) Non-price competition would not be involved in the allocation of the tickets under the system.

- A. (1) only
- B. (2) only
- C. (3) only
- D. (1), (2) and (3)

4. Which of the following is a normative statement / are normative statements?

- (1) Launching consumption voucher schemes will lead to inflation.
- (2) Promoting tourism is the most appropriate way to stimulate economic growth.
- (3) The government should continue to attract Mainland enterprises to invest in Hong Kong.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

5. Which of the following statements about the classification of goods is correct?

- A. Musical instruments used by street performers are public goods because the music performed with these instruments can be enjoyed concurrently by many people.
- B. Textbooks are private goods because the marginal cost of producing an additional unit of textbooks is positive.
- C. Artworks exhibited in the Hong Kong Palace Museum (香港故宮文化博物館) are consumer goods because costs are involved in their production.
- D. Popular free online videos for learning English are economic goods because their quantities are not enough to satisfy human wants.

6. Which of the following would raise both the labour supply and average labour productivity of an economy?

- A. The government raises the amount of unemployment benefits.
- B. Free education is extended to university programmes.
- C. More firms install advanced production equipment.
- D. The quota of an immigration scheme of highly skilled individuals is increased.

7. The occupational mobility of labour of an economy will decrease if

- A. transport subsidies are provided to workers who live in remote districts.
- B. more jobs require specialised skills.
- C. the statutory retirement age is raised from 65 to 70.
- D. more job retraining courses are provided by the government free of charge.

8. Refer to the following information about Firm A and Firm B in Hong Kong.

	Firm A	Firm B
Number of owners	3	50
Issuing shares to the public	Impossible	Possible
Disclosure of financial information to the public	Not necessary	Necessary

Based on the above information, we can conclude that

- A. Firm B raises more capital than Firm A.
 - B. owners of both firms are required to obtain consent from other owners when transferring their ownership.
 - C. Firm A may enjoy a lower profits tax rate than Firm B.
 - D. Firm B is an independent legal entity while Firm A is not.
9. The following table shows the data of a price-taking garment factory. The fixed cost of the factory is \$10.

Output (units)	Total cost (\$)
6	48
7	63
8	80
9	99
10	120
11	143

Suppose the market price of garments rises from \$17 to \$19, and the fixed cost of the factory rises by \$2. As a result, the profit-maximising output of the factory would _____ while its profit would _____.

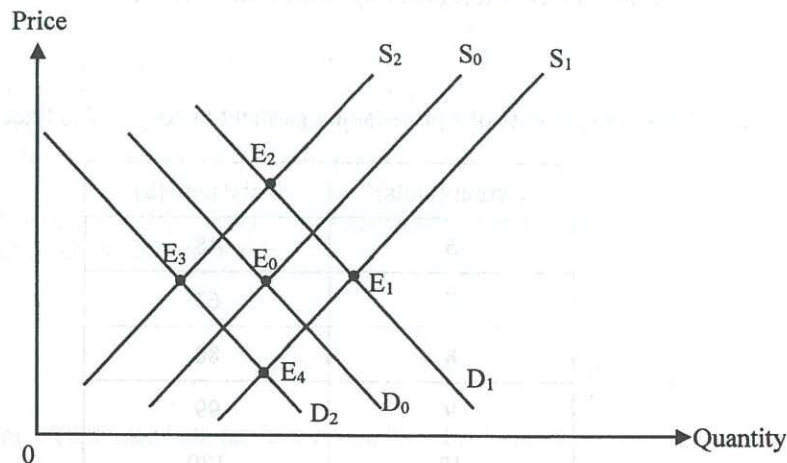
- A. remain unchanged increase to \$56
 - B. remain unchanged increase to \$70
 - C. increase by 1 unit increase to \$70
 - D. increase by 1 unit increase to \$82
10. Which of the following is **INCONSISTENT** with the law of demand?
- A. When a theme park reduces its ticket price, visitors stay in the theme park for longer hours on average.
 - B. Both the price and quantity transacted of private housing units decrease when people expect an increase in mortgage interest rates.
 - C. Compared with customers living close to supermarkets, customers living far away from supermarkets tend to buy more items in each visit to supermarkets.
 - D. If the Hong Kong Government shortens the quarantine period for incoming arrivals, the number of tourists visiting Hong Kong would increase.

11. When Good X and Good Y are in _____, a decrease in the price of Good X will lead to _____.

- (1) competitive demand a decrease in the price of Good Y
- (2) joint demand a decrease in the producer surplus in the market of Good Y
- (3) competitive supply an increase in the consumer surplus in the market of Good Y
- (4) joint supply a decrease in consumers' total expenditure on Good Y

- A. (1) and (3) only
- B. (1) and (4) only
- C. (2) and (3) only
- D. (2) and (4) only

12. The diagram below shows the supply and demand curves of electric vehicles. The original equilibrium point is E_0 .



Suppose there is an increase in the cost of producing electric vehicles. At the same time, the decreasing number of car accidents involving electric vehicles increases consumers' confidence in electric vehicles. As a result, the new equilibrium point will be at _____.

- A. E_1
- B. E_2
- C. E_3
- D. E_4

13. Semiconductor chips are key components for producing smartphones.

Due to a global shortage of semiconductor chips, the quantity of smartphones sold has changed by 10% whereas the price of smartphones has increased by 20%. Which of the following best explains the above changes in the price and quantity sold of smartphones?

- A. Smartphones have no close substitutes.
- B. Smartphone manufacturers have excess capacity in production.
- C. Consumers' expenditure on smartphones takes up a large proportion of their total expenditure.
- D. The workers for producing smartphones cannot be easily recruited.

14. Suppose the demand curve of face masks is a downward-sloping straight line. When the price of face masks falls from \$5 to \$4, consumers' total expenditure on face masks remains unchanged. If the price of face masks falls further from \$4 to \$3, consumers' total expenditure on face masks

A. will increase.
 B. will decrease.
 C. will remain unchanged.
 D. may increase, decrease or remain unchanged.

15. The initial equilibrium price of pineapples is \$5. Suppose the government imposes a per unit tax of \$4 on pineapples. If the elasticity of demand for pineapples is _____ than their elasticity of supply, the new equilibrium price can be _____.

(1) higher \$6
 (2) higher \$8
 (3) lower \$7
 (4) lower \$8

A. (1) and (3) only
 B. (1) and (4) only
 C. (2) and (3) only
 D. (2) and (4) only

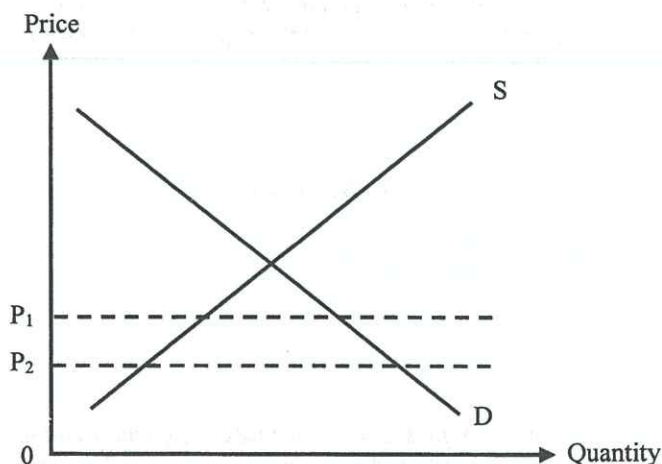
16. Refer to the following market demand and supply schedules of Good X.

Unit price (\$)	10	15	20	25	30	35
Quantity demanded (units)	140	120	100	80	60	40
Quantity supplied (units)	80	80	80	80	80	80

Which of the following statements about the market of Good X is correct?

A. If a price ceiling is set at \$10, the quantity transacted of Good X will be 140 units.
 B. If a production quota is set at 40 units, the price and quantity transacted of Good X will remain unchanged.
 C. If a per-unit subsidy of \$5 is provided, the equilibrium price of Good X will decrease to \$20.
 D. If a per-unit tax of \$5 is imposed, the net-of-tax revenue of Good X will decrease to \$1 600.

17. The following diagram shows the supply and demand curves of a good.



Suppose the government changes the price floor from P_1 to P_2 . Which of the following statements is/are correct?

- (1) The marginal benefit of the last unit of the good will decrease.
- (2) There will be an increase in the excess demand for the good.
- (3) Consumers' total expenditure on the good will decrease.
- (4) The quantity transacted of the good will remain unchanged.

- A. (4) only
- B. (1) and (4) only
- C. (2) and (3) only
- D. (1), (2) and (3) only

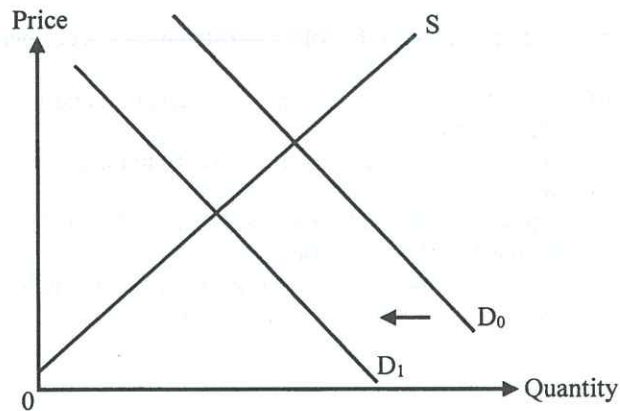
18. The table below shows the distribution of market share of Mandatory Provident Fund (強制性公積金) trustees in Hong Kong in 2021.

Mandatory Provident Fund trustee	Market share
Manulife	25.8%
HSBC	18.6%
Sun Life	10.4%
AIA	9.1%
Others	36.1%

Based on the above information, we can conclude that the above market is an example of _____ because _____.

- A. oligopoly sellers are interdependent in their marketing strategies
- B. oligopoly there are a few dominant sellers in the market
- C. monopolistic competition there are many sellers in the market
- D. monopolistic competition services provided by different sellers are heterogeneous

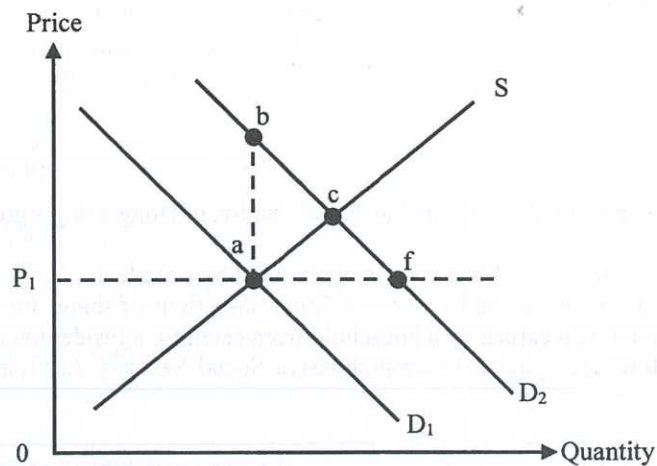
19. Refer to the following diagram about the market of local vegetables in an economy.



The shift of the demand curve from D_0 to D_1 _____.

- A. can be caused by an increase in the price of local vegetables
 - B. will raise the marginal cost of the last unit of local vegetables
 - C. will lead to economic inefficiency
 - D. can be caused by a decrease in the sales tax on imported vegetables
20. The provision of a per-unit production subsidy on a good will lead to _____.
- A. a fall in the producer surplus of the good
 - B. a rise in consumers' total expenditure on the good
 - C. a rise in the total social surplus of the good
 - D. a rise in the total revenue including the subsidy of the good

21. The following diagram shows the supply and demand curves of Good Z.



Suppose a price ceiling is set at P_1 . When the demand for Good Z increases from D_1 to D_2 , there is _____ and the deadweight loss is _____.

- A. over-production area abc
- B. over-production area acf
- C. under-production area abc
- D. under-production area acf

22. Plastic straws are a major source of plastic waste. Most plastic straws would break into smaller particles, releasing chemicals which threaten the environment.

Based on the above information, which of the following statements is correct?

- A. The use of plastic straws results in negative externality, and the quantity of plastic straws used is below the efficient level.
- B. To maximise total social surplus, governments around the world should completely ban the use of plastic straws.
- C. Charging a lump-sum tax on the businesses that provide plastic straws to customers must restore efficiency in the market of plastic straws.
- D. Granting a per-unit subsidy to the producers of environmentally-friendly paper straws tends to reduce the deadweight loss in the market of plastic straws.

23. Study the following data about an economy.

	2016	2021
Gini coefficient based on post-tax post-social transfer household income	0.398	0.407

Which of the following are possible explanations for the above change from 2016 to 2021?

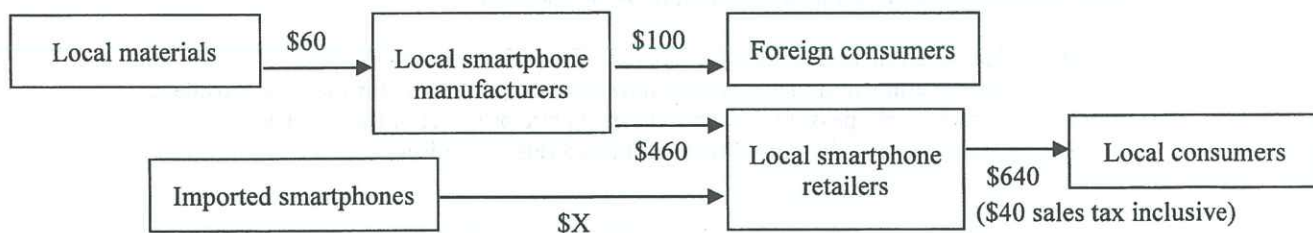
- (1) The progressive tax system changed to a regressive one.
- (2) The tax bands of the progressive salaries tax are widened.
- (3) The transfer payments provided to the low-income households increased.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

24. Which of the following would be included in the calculation of Hong Kong's gross domestic product (GDP)?

- A. the value of a second-hand bicycle purchased by a student
- B. the commission earned by a broker from transactions of shares for his clients
- C. the capital gain earned by a household from reselling a residential property
- D. the allowance under the Comprehensive Social Security Assistance Scheme received by the needy

25. The following diagram shows a production chain of smartphones in an economy.



If the contribution of the above production chain to the economy's GDP at factor cost is \$560, the value of X is _____.

- A. 40
 - B. 80
 - C. 140
 - D. 180
26. The following table shows some data of an economy in a certain year.

Factor income from abroad	\$14 billion
Factor income paid abroad	\$98 billion
Direct taxes	\$95 billion
Indirect taxes	\$23 billion
Production subsidies	\$57 billion
GDP deflator	117

Which of the following statements about the economy are correct?

- (1) The GDP is higher than the gross national income (GNI).
 - (2) The GDP at factor cost is higher than the GDP at market price.
 - (3) The nominal GDP is higher than the real GDP.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)
27. Study the following data of a country.

Year	2020	2021
Unemployed population (million)	176.9	175.4
Unemployment rate (%)	3.8	4.0

Which of the following best explains the above changes of the country from 2020 to 2021?

- A. More people look for jobs.
- B. More employed persons emigrate to other countries.
- C. The government raises the effective minimum wage rate.
- D. The government raises the amount of unemployment benefits.

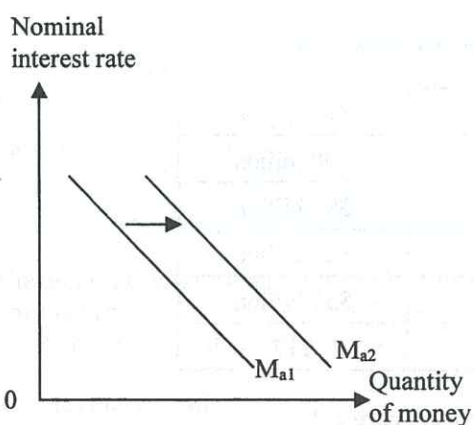
28. Suppose there is a fall in the national income of a country's major trading partner. Which of the following would offset its effect on the aggregate output of the country?

- A. The government increases the amount of tax rebate on salaries tax.
- B. The government imposes stricter quarantine requirements for inbound travellers.
- C. Firms become pessimistic about the economic prospect of the country.
- D. Landlords raise the rent of their office and retail premises.

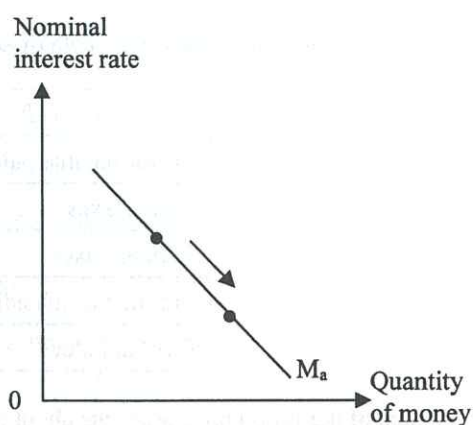
29. Suppose the stock market becomes more volatile. Which of the following diagrams best illustrates the effect of the above change on the money market?

M_a = asset demand for money
 M_t = transaction demand for money

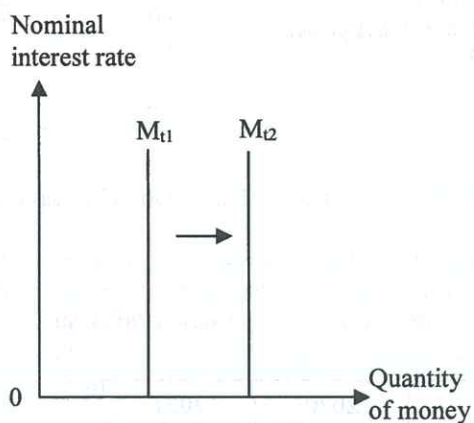
A.



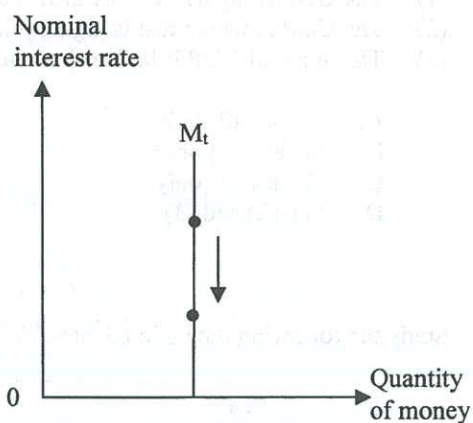
B.



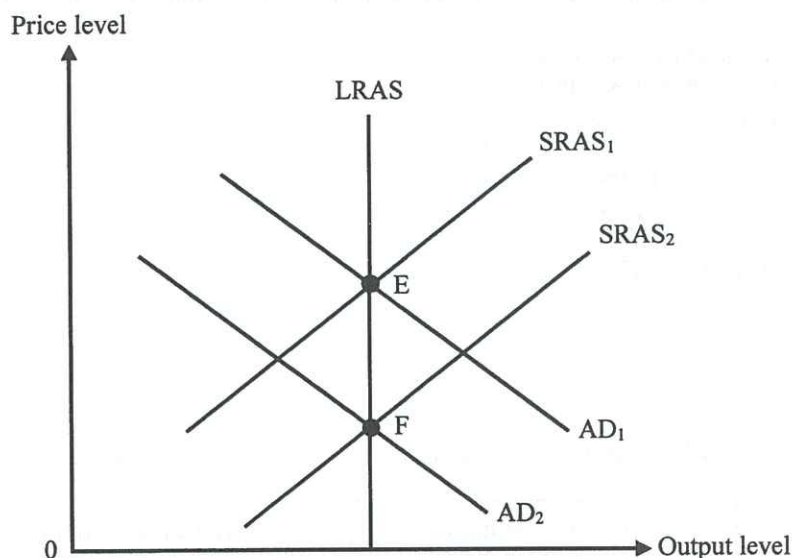
C.



D.



30. The diagram below shows the aggregate demand (AD) curves, the short run aggregate supply (SRAS) curves and the long run aggregate supply (LRAS) curve of an economy. The economy initially operates at point E.



Which of the following will result in a change in the long run equilibrium point from E to F?

- A. Households' desire to save increases.
- B. Prices of factor inputs decrease.
- C. The government offers more retraining programmes to workers.
- D. The required reserve ratio decreases.

31. Which of the following is a function of the Hong Kong Monetary Authority?

- A. providing loans to licensed banks and the public
- B. managing foreign exchange reserves
- C. determining the market interest rates
- D. collecting taxes for the government

32. Which of the following will lead to an increase in monetary base?

- A. The central bank raises the discount rate.
- B. The central bank reduces the required reserve ratio.
- C. The central bank buys government bonds from the public.
- D. The central bank replaces some of the paper banknotes in circulation with polymer banknotes.

33. Carmen withdraws HK\$600 000 from her savings deposits with a licensed bank in Hong Kong. She puts one-third of the amount into her current account and remits the remaining amount overseas.

What will be the immediate effects of the above actions on the Hong Kong dollar money supply?

- (1) M1 increases by HK\$200 000.
- (2) M2 decreases by HK\$600 000.
- (3) M3 decreases by HK\$400 000.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

34. Refer to the following balance sheet of a banking system which has excess reserves of \$125 million.

Assets (\$ million)		Liabilities (\$ million)	
Reserves	500	Deposits	1 500
Loans	1 000		

Which of the following statements about the above banking system is correct?

- A. The actual banking multiplier is 4.
- B. If the banking system loans out all the excess reserves, the amount of reserves will decrease after the deposit creation process is completed.
- C. If the banking system loans out all the excess reserves, the maximum possible increase in loans will be \$500 million.
- D. If the public withdraws \$125 million from the banking system and holds it as cash, the amount of excess reserves will decrease to zero immediately.

35. Bobby lives in his newly bought property in Wan Chai. He gains \$100 000 from trading shares in the stock market of Hong Kong.

Based on the above information, Bobby needs to pay _____ to the Hong Kong Government.

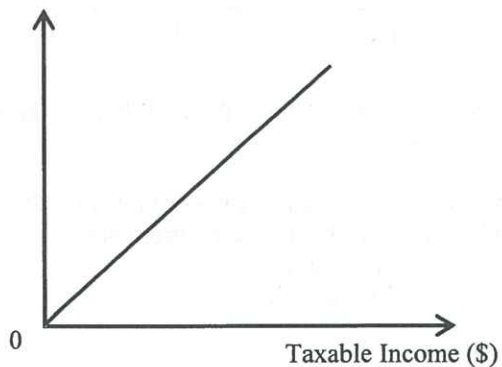
- (1) capital gains tax
- (2) property tax
- (3) rates
- (4) stamp duty

- A. (1) and (3) only
- B. (2) and (4) only
- C. (3) and (4) only
- D. (1), (2) and (3) only

36. Which of the following diagrams describe a progressive tax?

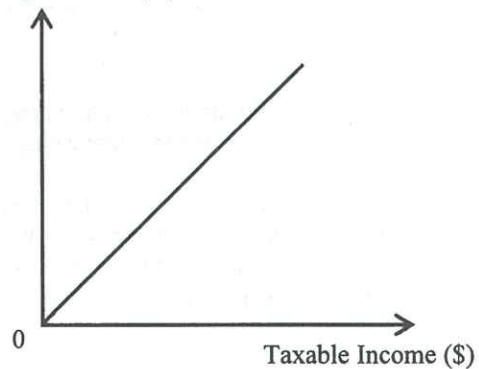
(1)

Tax payment (\$)



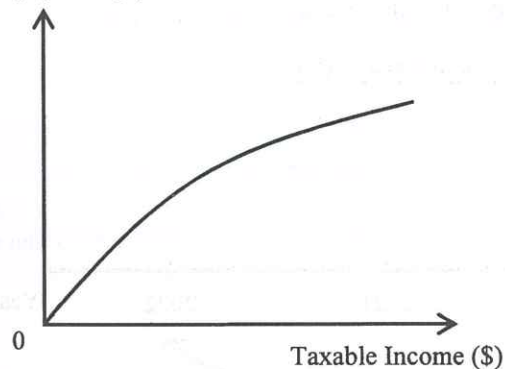
(2)

Average tax rate (%)



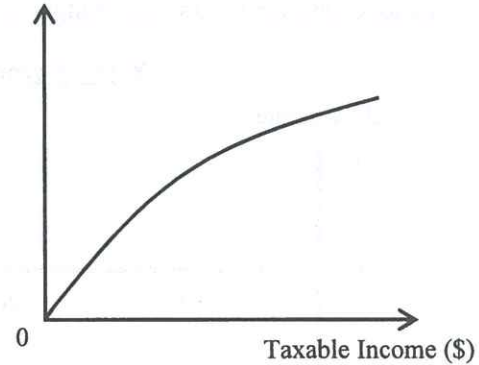
(3)

Tax payment (\$)



(4)

Average tax rate (%)



- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (4) only
- D. (3) and (4) only

37. According to the classical quantity theory of money, if the money supply of an economy increases by 8%,

- A. the government's income tax revenue will increase by 8%
- B. the nominal output will increase by 8%
- C. the real output will increase by 8%
- D. the price level may increase by less than 8%

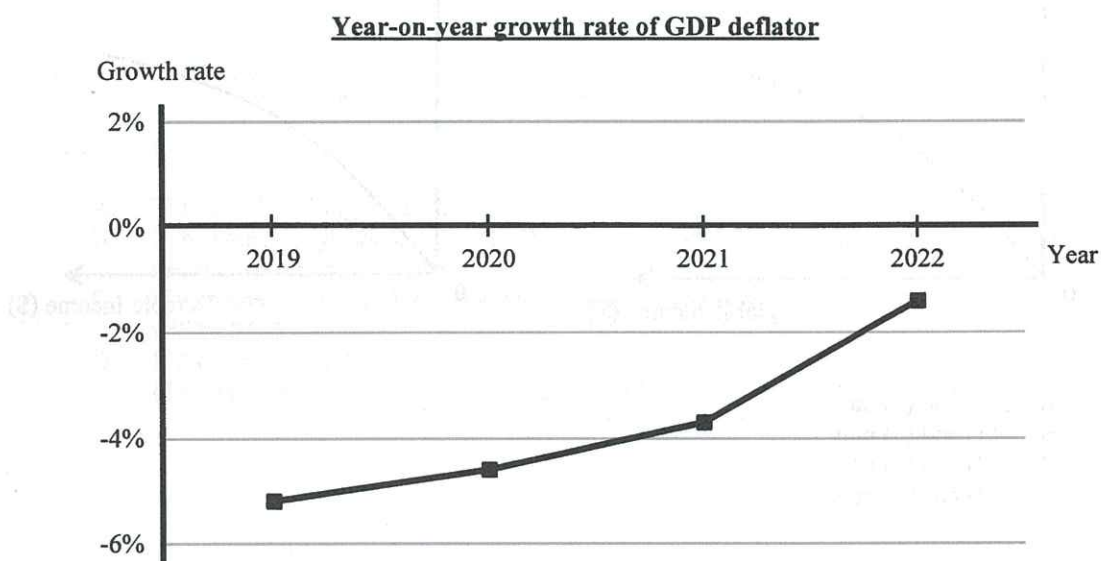
38. The table below shows the Composite Consumer Price Index of Hong Kong.

Year	Composite Consumer Price Index
2020	99.9
2021	101.4

Suppose people did not expect any change in the price level from 2020 to 2021. Which of the following persons would gain when the above change in price level occurred?

- A. a lender who provided a one-year loan to an enterprise at a fixed interest rate in 2020
- B. an employee whose salary in 2021 was adjusted according to the inflation rate
- C. a student who paid a fixed tuition fee in both 2020 and 2021
- D. an elderly who held time deposits in a bank from 2020 to 2021

39. The following chart shows the changes in the GDP deflator of a country.



Based on the above chart, which of the following statements about the country during the above period are correct?

- (1) The purchasing power of money increased.
 - (2) The nominal GDP growth rate was lower than the real GDP growth rate.
 - (3) The cost of living increased.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

40. Which of the following would result in a rise in the price level of an economy and a fall in its employment level at the same time?

- A. The government reduces the profits tax rate.
- B. The economy's currency appreciates against the currencies of its trading partners.
- C. The government increases its spending on research and development.
- D. The prices of imported raw materials increase.

41. Which of the following are examples of an expansionary fiscal policy?

- (1) increasing the child allowance of salaries tax
- (2) launching a consumption voucher scheme
- (3) lowering required reserve ratio

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

42. The following table shows the amount of apples or peaches that Country A and Country B can produce with one unit of resources.

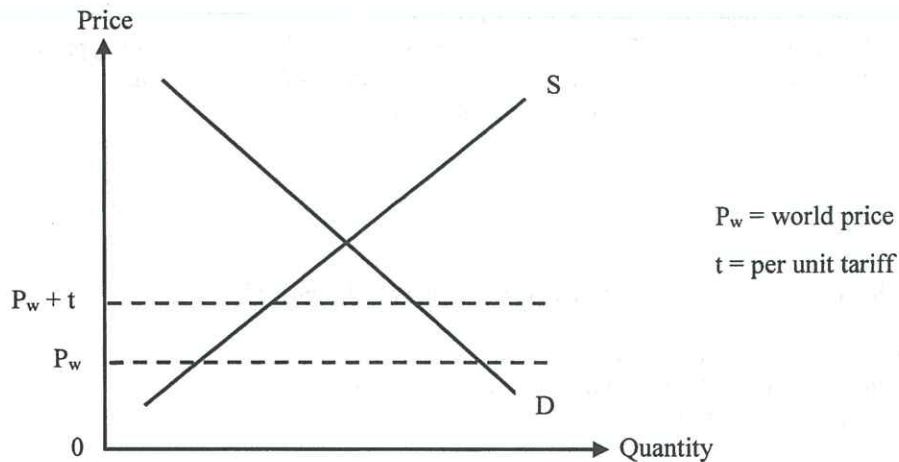
	Apples (units)		Peaches (units)
Country A	100	Or	80
Country B	180	Or	90

According to the above table, which of the following statements are correct?

- (1) The opportunity cost of producing 1 unit of apples is 0.8 units of peaches in Country A.
- (2) Country B has an absolute advantage in producing apples.
- (3) A mutually beneficial terms of trade can be 1 unit of apples = 0.7 units of peaches.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

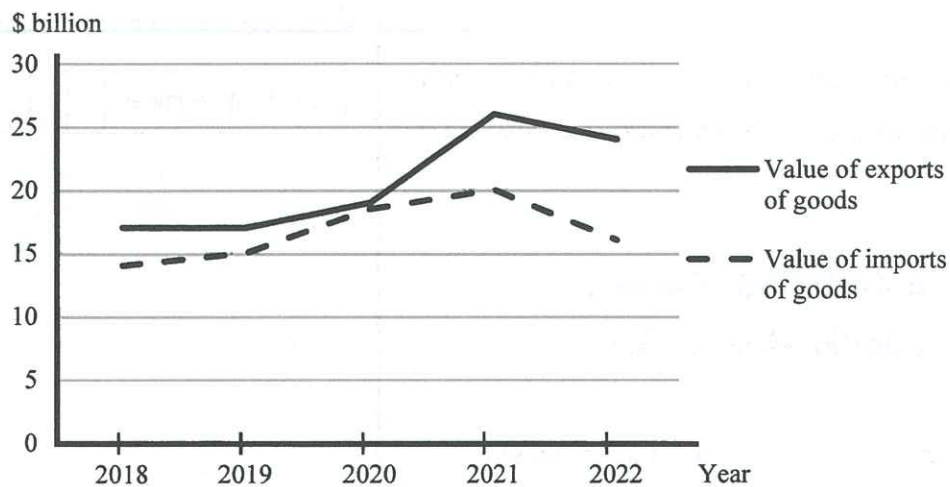
43. The diagram below shows the market of Good X in a small open economy. The economy has imposed a per unit tariff on Good X.



If the world price of Good X decreases, _____.

- A. the tariff revenue of Good X will remain unchanged
 - B. the domestic price of Good X will increase
 - C. the quantity of Good X consumed domestically will remain unchanged
 - D. the quantity of Good X produced domestically will decrease
44. Which of the following is included in the calculation of the balance of Hong Kong's current account?
- A. A Mainland billionaire spends HK\$20 billion building a shopping mall in Hong Kong.
 - B. A Japanese resident buys an air ticket to Hong Kong from an airline company in Japan.
 - C. A Hong Kong resident receives interest from US Government bonds.
 - D. A Hong Kong investor buys shares issued by a German listed company.

45. The figure below shows the values of a country's exports and imports of goods from 2018 to 2022.



From year _____ to year _____, the country had an increase in the value of imports of goods and an increase in visible trade surplus at the same time.

- A. 2018 2019
- B. 2019 2020
- C. 2020 2021
- D. 2021 2022

END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

Please stick the barcode label here.

Candidate Number

ECONOMICS PAPER 2
Question-Answer Book

10:15 am – 12:45 pm (2 hours 30 minutes)
This paper must be answered in English

INSTRUCTIONS

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7, 9 and 11.
- (2) This paper consists of two sections, A and B. Answer ALL questions in sections A and B.
- (3) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (4) Supplementary answer sheets will be supplied on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string INSIDE this book.
- (5) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.



Section A (44 marks)

1. An electric car manufacturer took over a social media company.

- (a) Explain which type of expansion this was. (2 marks)
- (b) Why did the electric car manufacturer choose to take over a social media company instead of taking over another electric car manufacturer? List **TWO** possible reasons. (2 marks)
- (c) Some economists share their economic analyses on social media free of charge. Explain whether these economic analyses are free goods to society. (2 marks)

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2. Study the following news extract.

A postal company has decided to introduce its first digital employee, a smart chatbot (智能聊天机器人). The chatbot can answer commonly asked questions from customers, freeing up the customer services team to solve more complex queries.

- (a) Explain to which type of factor of production the chatbot belongs. (2 marks)
- (b) Which of the three basic economic problems is the decision of the company related to? Explain your answer. (2 marks)

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Answers written in the margins will not be marked.

3. The table below shows the production data of a firm.

Machinery (units)	Labour (units)	Average output of labour (units)
12	10	10
12	11	12
12	12	15
12	13	17
12	14	16

- (a) Based on the above data, explain why machinery is a fixed factor. (1 mark)
- (b) Explain whether the law of diminishing marginal returns is illustrated by the above data. (3 marks)

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4. Smartphones of the same model are sold in different shops at different prices. Explain with **TWO** reasons why some customers buy the smartphones from shops selling at higher prices. (4 marks)

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2023-DSE-ECON 2-5

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6. Study the following statistical data of Hong Kong.

Year	Labour Force (thousand persons)
2020	3 918.5
2021	3 870.4

'The above change in labour force necessarily implies a rise in the unemployment rate in 2021 in Hong Kong'.
Do you agree? Explain your answer. (4 marks)

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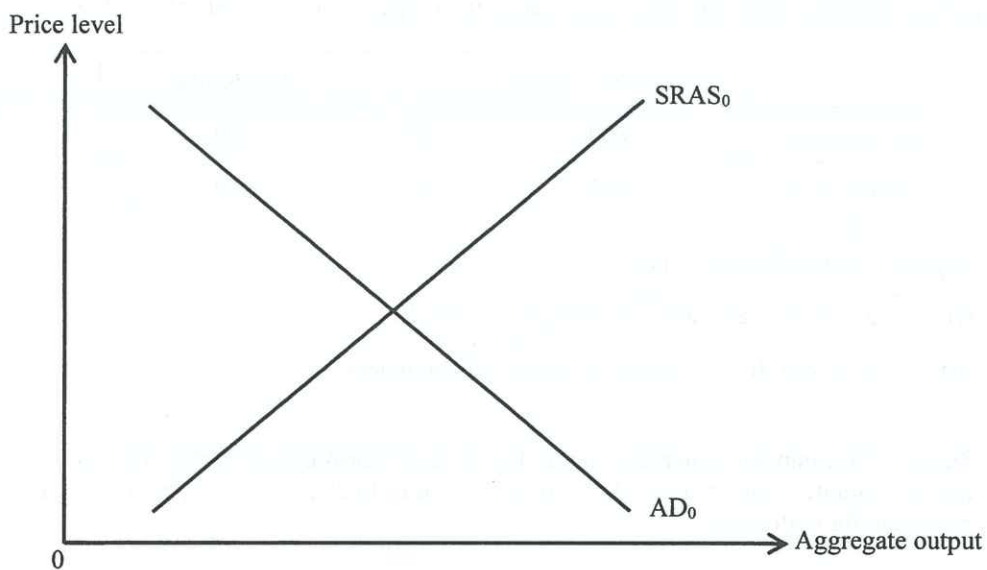
7. There exists a deflationary (output) gap in a country.

(a) Indicate the deflationary (output) gap in Figure 1.

(1 mark)

(b) To attract travellers, the government relaxes the quarantine requirements for inbound tourists. With the aid of Figure 1, explain how the deflationary (output) gap would be affected. (5 marks)

Figure 1



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8. The amount of resources in Country A is twice of that in Country B. Suppose both Country A and Country B use **ALL** their resources in production. Their maximum output is shown in the table below.

	Computers (units)		Rice (units)
Country A	800	Or	500
Country B	400	Or	300

- (a) Explain whether Country A has
- (i) an absolute advantage in producing computers.
 - (ii) a comparative advantage in producing computers.
- (5 marks)
- (b) Suppose the computer-importing country has to pay a transportation cost of 0.05 units of rice for each unit of computers imported. Find the range of terms of trade (for 1 unit of computers) that is mutually beneficial for both countries.
- (2 marks)

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Answers written in the margins will not be marked.

9. Figure 2: Value of US dollar (in terms of Japanese yen)

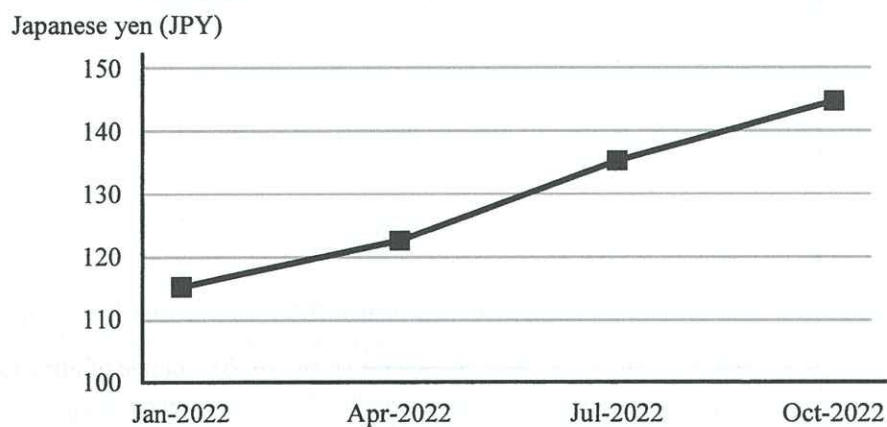
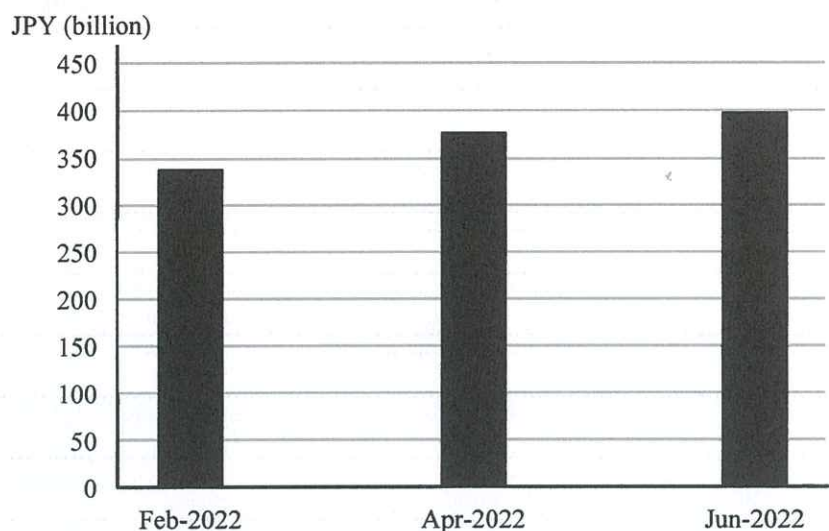


Figure 3: Total value of Hong Kong's imports from Japan (in terms of JPY)

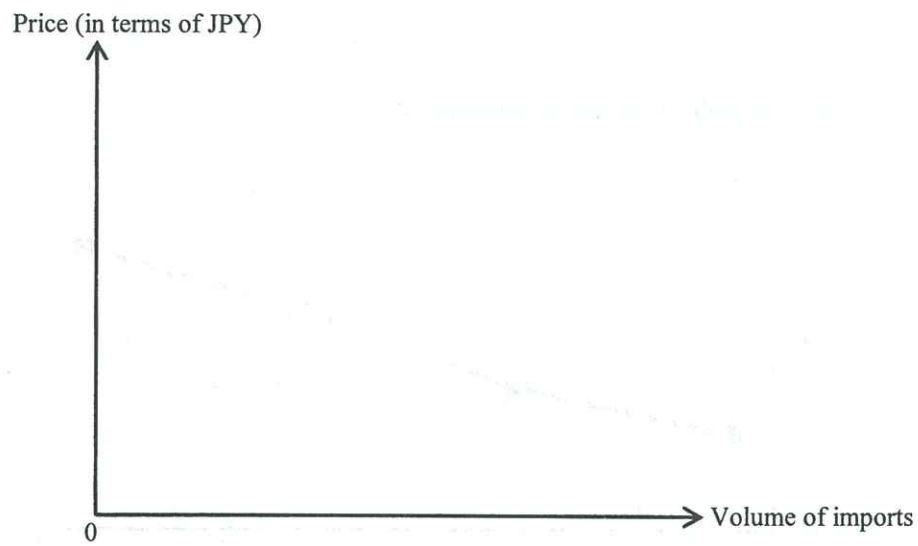


- (a) Refer to Figure 2. Under the Linked Exchange Rate System of Hong Kong, how would the exchange rate between JPY and Hong Kong dollar (HKD) change between January and October 2022? (1 mark)
- (b) 'The change in the total value of imports shown in Figure 3 could be explained by the change in exchange rate between JPY and HKD during the period.' With the aid of Figure 4, justify this statement.
(For simplicity, assume that in terms of JPY, the price of Hong Kong's imports from Japan does not change.) (4 marks)

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Figure 4



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END OF SECTION A

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Section B (60 marks)

10. In 2022, the Hong Kong Financial Secretary announced the issuance of Silver Bonds for citizens aged 60 or above to subscribe. The nominal interest rate of the Silver Bonds is equal to the actual inflation rate in Hong Kong, subject to a minimum of 4%.
- (a) From the viewpoint of senior citizens, explain **ONE** advantage and **ONE** disadvantage of holding Silver Bonds over holding cash. (4 marks)
- (b) Suppose the expected inflation rate is 4.5% and there turns out to be an actual deflation of 1%.
- (i) Calculate the expected and the actual real rates of return on the Silver Bonds. (2 marks)
- (ii) Based on your answer in (b)(i), explain whether holders of the Silver Bonds would gain or lose. (2 marks)
- (c) Suppose a large number of senior citizens withdraw money from their bank accounts to purchase Silver Bonds.
- (i) Describe the process of deposit creation/contraction which may result from the above withdrawal. (3 marks)
- (ii) Explain with **ONE** reason why the above withdrawal does not necessarily lead to deposit creation/contraction in reality. (2 marks)
- (d) The government issued \$45 billion worth of Silver Bonds in 2022. Explain whether this \$45 billion was included in the calculation of Hong Kong's GDP in 2022. (2 marks)
- (e) The Financial Secretary is paid a fixed monthly salary.
A teacher says, 'It is impossible to pay the Financial Secretary with a piece rate.' Justify the teacher's statement. (2 marks)

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11. In Hong Kong, fans of popular singers often complain that they are unable to buy concert tickets of their idols at the official prices.

- (a) Figure 5 shows the demand for and the supply of the concert tickets of a popular singer in Hong Kong. With the aid of Figure 5, explain the above phenomenon. (4 marks)

Some fans who are unable to purchase concert tickets through official channels would choose to buy tickets from touts (黃牛黨) at prices substantially higher than the official prices.

- (b) Which function of price is illustrated in the above situation? Explain your answer. (2 marks)

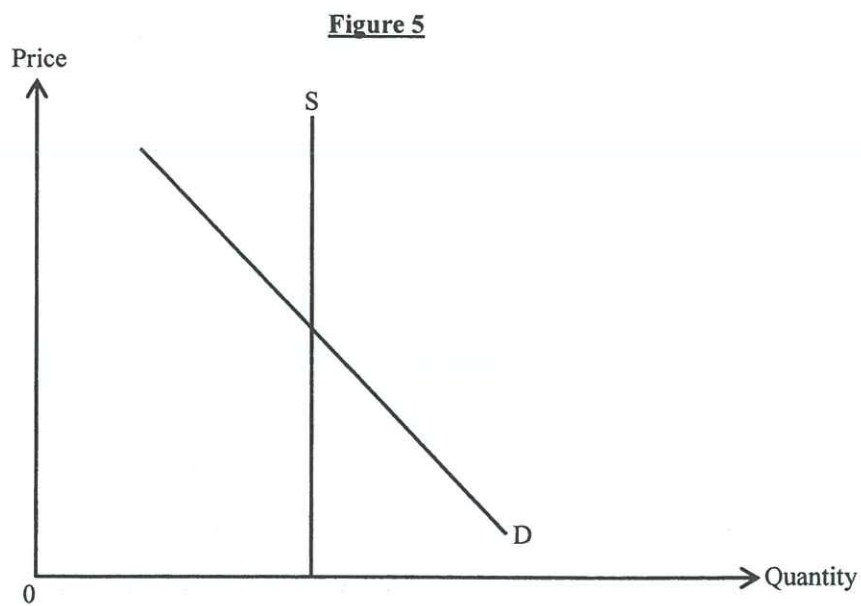
Below is an extract of some interviews about a concert.

a housewife 	I queued up at the official ticketing office for 10 hours as expected. I was finally able to buy a ticket!
a concert organiser 	Had we known that the fans were so supportive, we would have held more shows for the concert. This would help reduce the average cost per show.

- (c) Refer to the interview of the housewife. Explain, in terms of opportunity cost, why she chose to buy the concert ticket at the official ticketing office instead of buying from touts. (3 marks)
- (d) Refer to the interview of the concert organiser. Explain with **TWO** reasons why holding more shows for the same concert may help reduce the average cost per show. (4 marks)
- (e) It is observed that fans from overseas tend to buy higher-priced concert tickets when buying through the official channel. Based on the law of demand, explain this phenomenon. (2 marks)

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12. Inflation is a concern in the US. Different causes of inflation have been suggested, including rising crude oil prices, expansionary monetary policy, and high import tariffs.

Source A: Prices of crude oil from May 2020 to May 2022



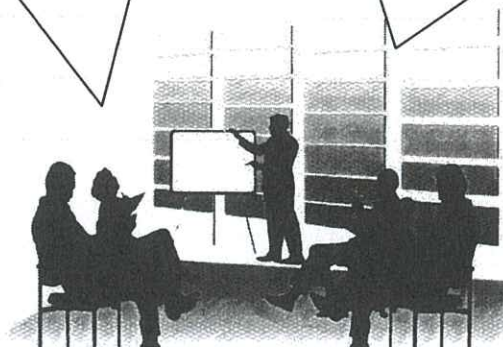
Source B: Some views about energy prices expressed by participants in a forum

A representative of manufacturers:

Changes in crude oil prices affect us a lot. Many of our products such as gasoline and plastics are made from crude oil.

A typical US citizen:

We need gasoline for our daily routines such as driving to work. Although we try very hard to reduce gasoline consumption, our bills for gasoline are still getting higher and higher.

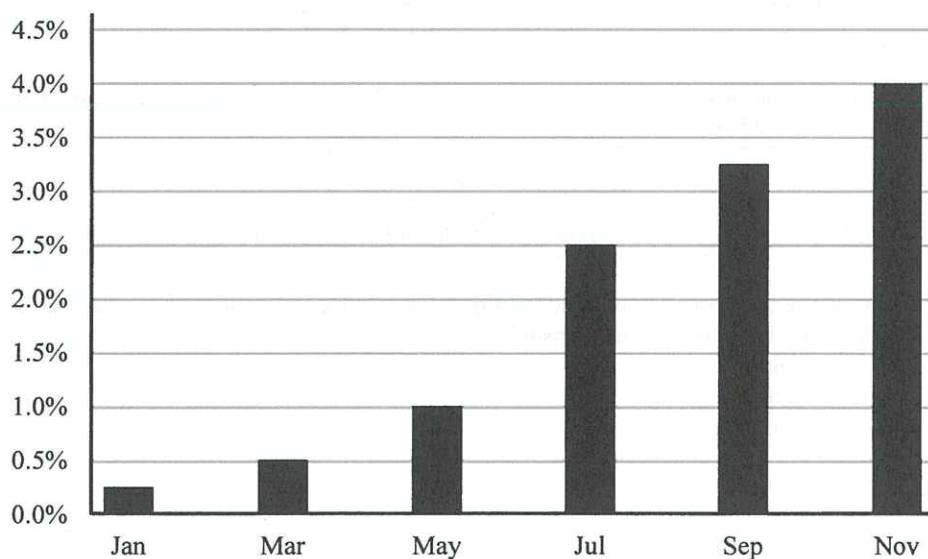


Source C: Money supply in the US

	May 2020	May 2021
Money supply (USD Billion)	17 766.9	20 319.6

Answers written in the margins will not be marked.

Source D: Discount rate of the Federal Reserve* of the US from January to November 2022



* Federal Reserve is the central bank of the US.

Source E: Some comments on the US tariffs on Chinese imports

A US producer 	We can no longer bear the costs of tariffs on raw materials imported from China without increasing our product prices.
An economist 	The tariffs cause the prices of imported consumer goods from China to rise. This puts a greater burden on low-income households than on high-income households.

- (a) Refer to Sources A and B.
- (i) Explain how the change in crude oil prices shown in Source A affects gasoline prices. (3 marks)
- (ii) With reference to your answer in (a)(i), explain why consumers' total expenditure on gasoline would increase. (3 marks)
- (b) Refer to Source C. According to the quantity theory of money, explain under what condition the change in money supply as shown in the source would result in an increase in the general price level in the short run. (3 marks)
- (c) Refer to Source D. How may the change in discount rate as shown in the source help relieve the pressure on rising price level? Without using a diagram, explain your answer in the light of aggregate supply and/or aggregate demand. (5 marks)

Answers written in the margins will not be marked.

For part (d), candidates are required to present their answers in an essay form. Criteria for marking will include the use of sources and economic theories, relevant content, logical presentation and clarity of expression.

- (d) Apart from adjusting the discount rate, two other strategies have been suggested to address the problem of inflation in the US.

Strategy 1	Raising the salaries tax rate for the high-income group
Strategy 2	Cancelling the tariffs on Chinese imports

With reference to Sources A to E and your own knowledge in Economics, discuss the effects of **EACH** strategy on the US economy in terms of

- price level,
- equity, and
- trade balance.

(14 marks)

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END OF PAPER

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Answers written in the margins will not be marked.

Supplementary Answer Sheet

Question No. _____

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Answers written in the margins will not be marked.

Marking Schemes

Paper 1

Question No.	Key	Question No.	Key
1.	C (84)	26.	D (47)
2.	A (82)	27.	B (71)
3.	B (80)	28.	A (72)
4.	D (61)	29.	A (67)
5.	D (63)	30.	A (57)
6.	D (78)	31.	B (76)
7.	B (93)	32.	C (66)
8.	C (59)	33.	B (73)
9.	C (67)	34.	C (58)
10.	A (64)	35.	C (47)
11.	A (48)	36.	C (56)
12.	B (89)	37.	B (61)
13.	A (72)	38.	C (73)
14.	B (17)	39.	A (51)
15.	B (62)	40.	D (59)
16.	D (50)	41.	A (51)
17.	A (64)	42.	D (62)
18.	B (76)	43.	D (58)
19.	D (62)	44.	C (53)
20.	D (47)	45.	C (77)
21.	C (60)		
22.	D (56)		
23.	A (77)		
24.	B (80)		
25.	C (44)		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

Paper 2

This document was prepared for markers' reference. It should not be regarded as a set of model answers. Candidates and teachers who were not involved in the marking process are advised to interpret its content with care.

The answers provided in the marking scheme are for reference only. They are not the only possible answers. Alternative answers are acceptable so long as they are well reasoned.

The examination emphasises the testing of the understanding of economic theories and the application of the knowledge of economic analysis to practical problems. Candidates are advised to study this document in conjunction with the examiner's comments on candidates' performance in this booklet.

For essay-type questions, candidates are expected to demonstrate an understanding of the question, an ability to deploy relevant knowledge of the subject in response to the questions, and to present their answers logically and coherently.

In questions asking for a specified number of reasons or examples etc. and a candidate gives more than the required number, the extra answers should not be marked. For instance, in a question asking candidates to provide two examples, and if a candidate gives three answers, only the first two should be marked.

The following symbols are used:

- | | |
|-----|---|
| / | A single slash indicates an acceptable alternative within an answer. |
| @ | The number in front of the symbol indicates the marks for each point. |
| max | Maximum mark for the question/sub-question |

Section A

- | | | Marks |
|--|---|------------------|
| 1. | (a) Conglomerate integration, because electric car manufacturing and social media operation are totally unrelated businesses. | (1)
(1) |
| (b) <u>Possible motives:</u> <ul style="list-style-type: none">– to spread risk through product diversification– to make use of its brand name in other products/services– any other relevant point [Mark the FIRST TWO points only.] | | } 1 @;
max: 2 |
| (c) No, because the resources involved in writing these analyses have alternative uses.
OR
more of these analyses are preferred by some people. | | (1)
(1) |
| | | |
| 2. | (a) Capital, because it is a man-made resource used in production. | (1)
(1) |
| (b) 'How to produce', because replacing labour with capital is related to the production method / combination of factor inputs. | | (1)
(1) |

3. (a) Its quantity does not vary with output.

Marks
(1)

(b) Yes, because

the marginal product of labour drops from 48 to 41 units when the quantity of labour (a variable factor) increases from 12 to 13 units, given a fixed quantity (12 units) of machinery (fixed factor).

(1)
(2)

4. Possible reasons:

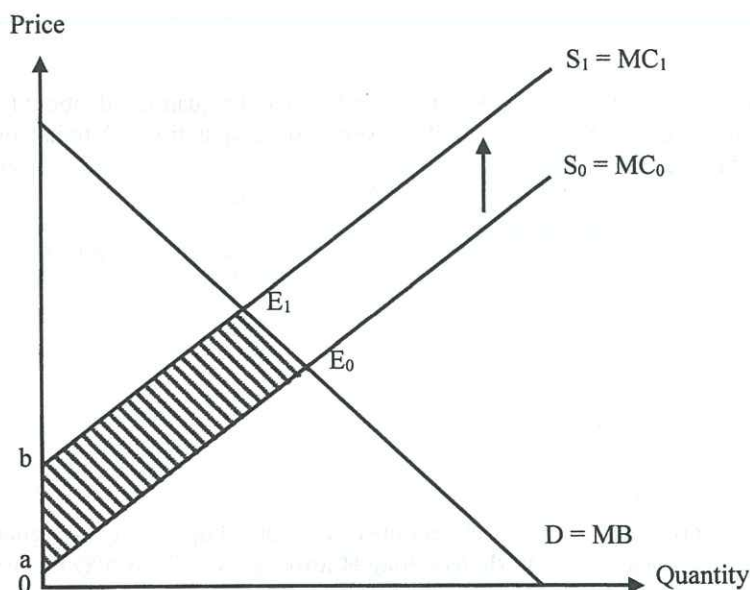
- Imperfect market information. Some customers may buy from shops charging higher prices because they do not know about shops selling at lower prices. (This may be due to high costs of acquiring such information, viz., search costs.)
 - Heterogeneous products/services. Some customers may be willing to pay a premium for better services, e.g. better after-sales services such as longer warranty period / better product descriptions by salespersons.
 - Any other reasonable answer
- [Mark the **FIRST TWO** points only.]

} 2@;
max: 4

Marks

5. (a) Indicate in the diagram:
 – upward shift of MC curve (leftward shift of supply curve)
 – decrease in total social surplus

(1)
(1)



Decrease in total social surplus = area (abE₁E₀)

- (b) Verbal elaboration:
 No, efficiency is still attained because
 at the new equilibrium quantity, marginal benefit is still equal to marginal cost.

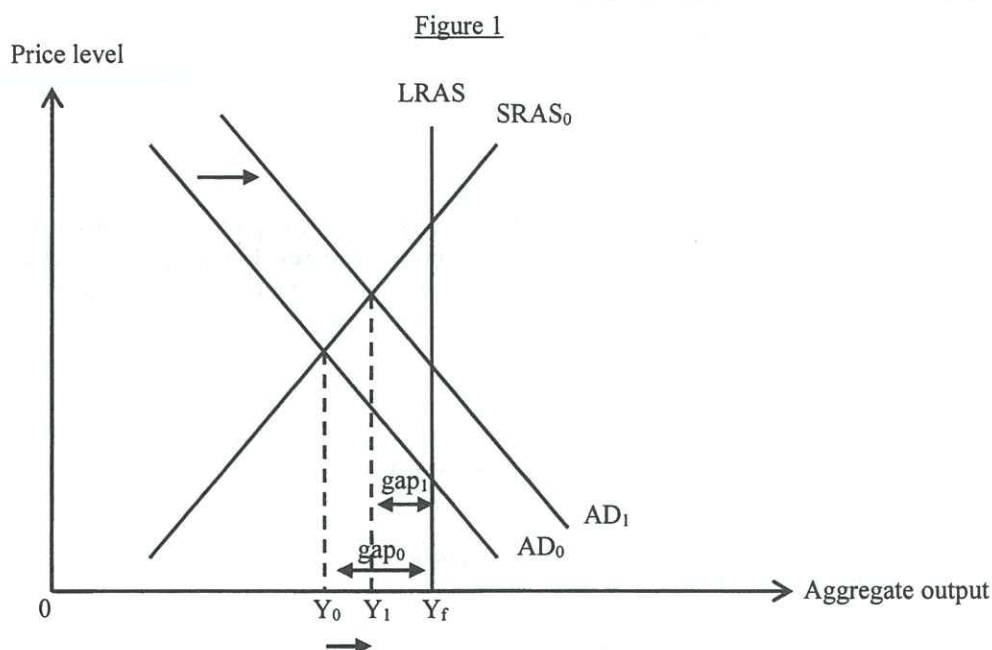
(1)
(1)

6. No. Unemployment rate = (unemployed population / labour force) × 100%.
 Despite the decrease in the size of HK's labour force, the unemployment rate would fall
 (or remain unchanged) if the percentage decrease in the unemployed population was
 greater than (or equal to) that in the labour force.

(2)
(2)

7. (a) Indicate in the diagram:
 – correct position of vertical LRAS curve and deflationary (output) gap

(1)



- (b) Verbal elaboration:

- As more travellers visit the country, export of services increases.
- Aggregate demand increases.
- Aggregate output increases, narrowing the deflationary (output) gap.

(1)

(1)

(1)

Indicate in the diagram:

- rightward shift of AD curve
- increase in Y, decrease in deflationary (output) gap

(1)

(1)

8. (a) (i) No, because with the same amount of resources, Country A can only produce the same number of computers as Country B.

(2)

- (ii) Yes.

Opportunity cost of producing 1 unit of computers:

in Country A = $500/800 = 0.625$ units of rice

in Country B = $300/400 = 0.75$ units of rice

As Country A can produce computers at a lower opportunity cost than Country B, Country A has a comparative advantage in producing computers.

(1)

(1)

(1)

- (b) 0.625 units of rice $<$ 1 unit of computers $<$ 0.7 units of rice

(2)

9. (a) Figure 2 shows that JPY depreciated against USD. Under the Linked Exchange Rate System, JPY would also depreciate against HKD.

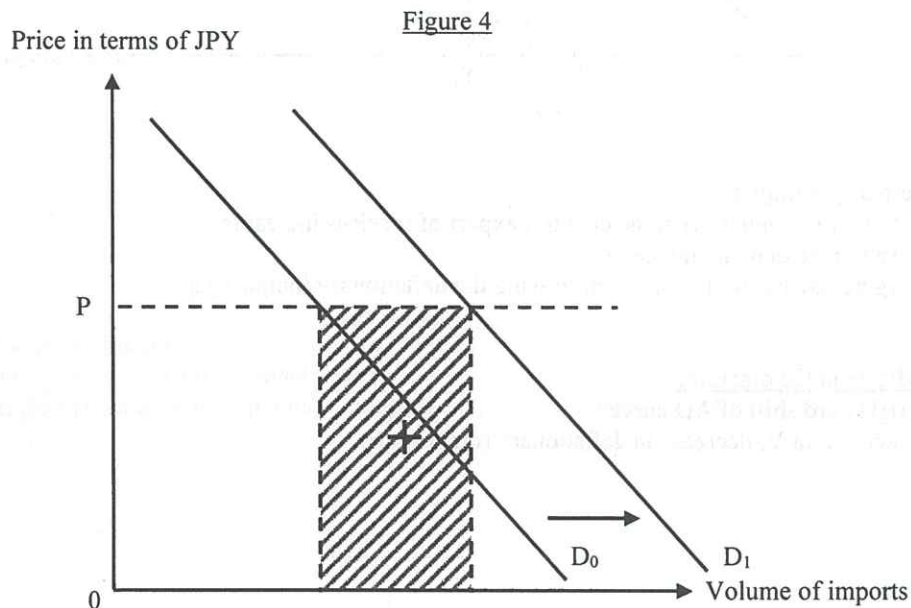
Marks
(1)

(b) Verbal elaboration:

- Import price in terms of HKD decreases. (1)
 - At any given price in terms of JPY, Hong Kong will import more from Japan, i.e. Hong Kong's demand for Japanese imports increases. Import volume increases. (1)
- Given that the price in terms of JPY remains unchanged, total import value in terms of JPY increases, as shown in the data.

Indicate in diagram:

- rightward shift of D curve (1)
- correct position of increase in total import value (+) (1)



Section B

Marks

10. (a) Advantage of holding Silver Bonds over cash:
Purchasing power can be better preserved in times of inflation since the nominal rate of return on Silver Bonds rises with inflation while that of cash remains at zero regardless of inflation rate. (2)
- Disadvantage of holding Silver Bonds over cash:
Liquidity is lower since it takes days for Silver Bonds to be redeemed and converted into cash. (2)
- (b) (i) Expected real rate of return = $4.5\% - 4.5\% = 0\%$ (1)
Actual real rate of return = $4\% - (-1\%) = 5\%$ (1)
- (ii) Gain, because (1)
the actual real rate of return is higher than the expected real rate of return. (1)
- (c) (i) The withdrawal of bank deposits may result in a reserve shortage. Banks would need to call back loans to fulfil legal reserve requirements. In order to keep a constant amount of cash (for daily expenses etc.), the public would have to further withdraw deposits from their bank accounts. The process would go on and on, resulting in deposit contraction. (3)
- (ii) Banks may have enough excess reserves to meet the withdrawal demands, so that there is no need to call back loans. (2)
- OR
The public may choose to reduce their cash holdings instead of further withdrawing bank deposits when their loans are being called back.
- (d) No, because (1)
no current production is involved. (1)
- (e) It is prohibitively costly to measure the outputs of the Financial Secretary as his outputs are not standardised. (2)

11. (a) Verbal elaboration:

- As the concert ticket price is set below the equilibrium price, there is an excess demand for concert tickets.

(2)

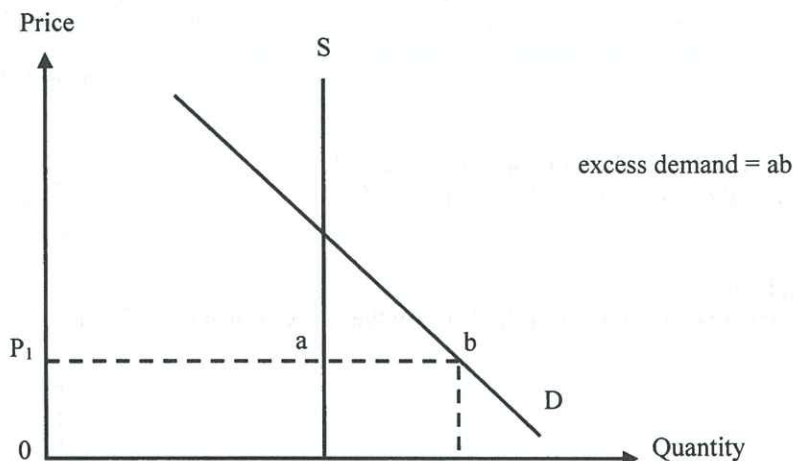
Indicate in the diagram:

- price below equilibrium price
- excess demand (ab)

(1)

(1)

Figure 5



- (b) rationing function: concert tickets are distributed to the highest-valued buyers.

(2)

- (c) Full cost = time cost + money cost.

(3)

The housewife chose to buy at the official ticketing office because her time cost of queuing (i.e. her value of the best alternative use of the queuing time) was smaller than the additional money cost of buying from touts (i.e. the difference between touting price and official price).

(d) Possible reasons:

- The cost of advertising can be spread over a larger quantity of outputs (number of shows).
- The cost of rehearsal can be spread over a larger quantity of outputs.
- any other relevant point

[Mark the **FIRST TWO** points only.]

} 2@;
max: 4

- (e) For fans from overseas, the same amount of travelling expenses is added to both higher-priced and lower-priced tickets.

(1)

Inclusion of these travelling expenses would lower the relative price of higher-priced tickets in terms of lower-priced tickets.

(1)

- | | Marks |
|---|--------------|
| 12. (a) (i) According to Source B, crude oil is an input for producing gasoline. As the prices of crude oil increase, the cost of producing gasoline increases. | (1) |
| Supply of gasoline decreases. | (1) |
| Price increases. | (1) |
| | |
| (ii) According to Source B, gasoline is a daily necessity. | (1) |
| Therefore, the demand for gasoline tends to be inelastic. | (1) |
| The percentage decrease in quantity demanded is smaller than the percentage increase in price. | (1) |
| Therefore, total expenditure on gasoline increases as its price increases. | |
| | |
| (b) Quantity Theory of Money: $MV = PY$ | (1) |
| M = money supply | |
| V = velocity of circulation of money | |
| P = price level | |
| Y = real output | |
| Assuming V to be constant, | (1) |
| as M increases, P increases if the growth rate of M exceeds that of Y. | (1) |
| | |
| (c) Increase in discount rate raises the cost for commercial banks to borrow money from the Federal Reserve. Monetary base tends to decrease. | (1) |
| Money supply decreases. | (1) |
| Interest rates increase. ¹ | (1) |
| Investment decreases. | (1) |
| Aggregate demand decreases. Price level tends to decrease, relieving the pressure on the rising price level. | (1) |

¹ Decrease in money supply would raise the nominal interest rate. If the change in money supply has little or no effect on the expected inflation rate, then Fisher's equation implies that the real interest rate would rise along with the nominal rate as well. The rise in real interest would then make it more costly for firms to borrow to finance their investment projects and for households to borrow to finance their consumption purchases. As a result, investment and consumption, and thus aggregate demand, would fall.

(d) Marks for effective communication (EC: max 2 marks)

Marks	Performance
2	● Support arguments with the source/data and appropriate economic theories. ● Present relevant material. ● Present well-organised and coherent answers without repetition of ideas. ● Use language that expresses ideas clearly and fluently with appropriate use of words/terms/symbols.
1	● Present arguments with some support of the source/data and economic theories. ● Present some irrelevant material. ● Present answers in a less organised way with some repetition. ● Use language that conveys a clear message with some inappropriate use of words/terms/symbols.
0	● Present arguments with no support of the source/data and economic theories. ● Present material unrelated to the gist of the question. ● Present inconsistent arguments. ● Express limited ideas with inappropriate use of words/terms/symbols.

The maximum mark for content is 12 marks

Answers may include the following:

- **Effects on price level**

Strategy I (raising the salaries tax rate for the high-income group)

- As the salaries tax rate for the high-income group is raised, their disposable income decreases. Consumption decreases. AD decreases. Price level decreases.
- As the salaries tax rate for the high-income group is raised, their working incentive decreases. Labour supply decreases. SRAS and/or LRAS decrease(s). Price level increases.

Strategy II (cancelling the tariffs on Chinese imports)

- As tariffs on imported inputs from China are cancelled, input prices decrease. Cost of production decreases. SRAS increases. Price level decreases.
- As tariffs on imported consumer goods from China are cancelled, their prices decrease. Import volume increases, implying a reduction in net exports. AD decreases. Price level decreases.

- **Effects on equity**

Strategy I (raising the salaries tax rate for the high-income group)

- As the salaries tax rate for the high-income group is raised, their disposable income decreases, while that of the low-income group is not affected. Therefore, this will help equalise (after-tax) income.

Strategy II (cancelling the tariffs on Chinese imports)

- When purchasing the same imported goods, the same amount of decrease in import prices caused by the cancellation of tariffs represents a greater proportion of income for the low-income group than the high-income group. Therefore, cancelling the tariffs will help equalise (real) income.

- **Effects on trade balance**

Strategy I (raising the salaries tax rate for the high-income group)

- As disposable income decreases, consumption, including the consumption of imported goods, decreases. Imports decrease. Given that the value of exports remains unchanged, trade balance of the US will improve.
- As AD decreases, price level decreases. Assuming that prices of other countries' products remain unchanged, prices of US products would decrease relative to goods produced in other countries. US exports tend to increase and US imports tend to decrease, thus improving the trade balance of the US.

Strategy II (cancelling the tariffs on Chinese imports)

- Cancellation of tariffs on imports from China would lower prices of those goods and thus increase the volume of imports from China. Given that the value of exports remains unchanged, the trade balance of the US will be worsened.
- As prices of imported raw materials from China decrease, costs of production decrease. SRAS increases. Price level decreases. Assuming that prices of other countries' products remain unchanged, prices of US products would decrease relative to goods produced in other countries. US exports tend to increase and US imports tend to decrease, thus improving the trade balance of the US.

Candidates' Performance

Paper 1

Candidates' overall performance was satisfactory. The mean score of this paper was 29 out of 45. The following questions deserve special attention.

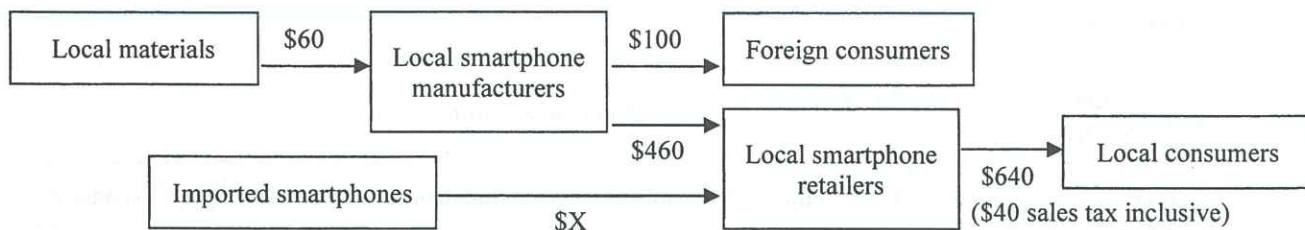
14. Suppose the demand curve of face masks is a downward-sloping straight line. When the price of face masks falls from \$5 to \$4, consumers' total expenditure on face masks remains unchanged. If the price of face masks falls further from \$4 to \$3, consumers' total expenditure on face masks
- A. will increase. (5%)
 - *B. will decrease. (17%)
 - C. will remain unchanged. (47%)
 - D. may increase, decrease or remain unchanged. (31%)

This question was related to the price elasticity of demand along a straight-line demand curve. Candidates picking 'C' as their answer might have either overlooked the words 'straight line' in the question and misconceived the demand curve as a rectangular hyperbola with a constant price elasticity of one (in absolute value), or misunderstood that the price elasticity is constant along a straight-line demand curve. It should be noted that on a straight-line demand curve, the price elasticity is greater than one (i.e., 'elastic') along any arc above its mid-point and smaller than one (i.e., 'inelastic') on any arc below the mid-point. The given information in the question implied that the mid-point of the demand curve was between \$5 and \$4. Hence, the arc of the demand curve between \$4 and \$3 lay below its mid-point and had a price elasticity less than one. With an inelastic demand, when the price fell from \$4 to \$3, the percentage increase in quantity demanded would be smaller than the percentage decrease in price, implying a fall in consumers' total expenditure.

20. The provision of a per-unit production subsidy on a good will lead to _____.
- A. a fall in the producer surplus of the good (12%)
 - B. a rise in consumers' total expenditure on the good (7%)
 - C. a rise in the total social surplus of the good (33%)
 - *D. a rise in the total revenue including the subsidy of the good (47%)

This question examined candidates' understanding of the effects of a per-unit production subsidy. Candidates picking 'C' as their answer might have the misconception that total social surplus is simply the sum of consumer surplus and producer surplus, which is not true when tax revenue or subsidy expenditure is involved. Although the subsidy could increase both the consumer surplus and the producer surplus, the government's subsidy expenditure was actually greater than the sum of the increases in consumer surplus and producer surplus. As the provision of subsidy led to an increase in output quantity beyond its efficient level, marginal cost exceeded marginal benefit for the last unit of production, resulting in a deadweight loss (i.e., a loss in total social surplus). The size of the deadweight loss was equal to the excess of the government's subsidy expenditure over and above the sum of the increases in consumer surplus and producer surplus.

25. The following diagram shows a production chain of smartphones in an economy.



If the contribution of the above production chain to the economy's GDP at factor cost is \$560, the value of X is _____.

- | | | |
|-----|-----|-------|
| A. | 40 | (11%) |
| B. | 80 | (35%) |
| *C. | 140 | (44%) |
| D. | 180 | (11%) |

This question tested candidates' ability to calculate the contribution of a given production chain to an economy's gross domestic product (GDP). Candidates who chose 'A' as their answer might have overlooked the output value of \$100 from 'local smartphone manufacturers' to 'foreign consumers', which should be included in the economy's GDP as an export. Candidates picking 'B' as their answer had likely misinterpreted the 'local materials' as 'imported materials' and deducted the value of \$60 in their calculations of the contribution of the production chain to the economy's GDP. Candidates picking 'D' as their answer might have failed to notice that the output value of \$640 from 'local smartphone retailers' was based on market prices (rather than factor costs) as it was sales tax inclusive. When computing the contribution to GDP at factor cost, the \$40 sales tax had to be deducted.

35. Bobby lives in his newly bought property in Wan Chai. He gains \$100 000 from trading shares in the stock market of Hong Kong.

Based on the above information, Bobby needs to pay _____ to the Hong Kong Government.

- | | | |
|-----|-----------------------|-------|
| (1) | capital gains tax | |
| (2) | property tax | |
| (3) | rates | |
| (4) | stamp duty | |
| A. | (1) and (3) only | (5%) |
| B. | (2) and (4) only | (34%) |
| *C. | (3) and (4) only | (47%) |
| D. | (1), (2) and (3) only | (15%) |

This question examined candidates' understanding of taxation in Hong Kong. Bobby needed to pay stamp duty (Item #4) as both his purchase of a residential property and his trading of shares were subject to stamp duty in Hong Kong. The major distractor in this question was property tax (Item #2). In Hong Kong, property tax is a tax on rental income. Property owners who live in their own premises and do not receive any rental income are not subject to property tax. Moreover, some candidates included capital gains tax (Item #1) in their answer. These candidates did not know that there is no capital gains tax in Hong Kong.

Paper 2

Section A

Question Number	Performance in General
1(a)	Good. Most candidates were able to correctly identify and explain the type of expansion.
1(b)	Satisfactory. Most candidates provided relevant reasons. However, some candidates did not realise that the question asked about specific motives behind the conglomerate integration. They provided general answers such as 'economies of scale', which could be enjoyed by taking over another electric car manufacturer (i.e., horizontal integration) as well.
1(c)	Excellent.
2(a)	Fair. A number of candidates wrongly identified the chatbot as 'labour'. Some candidates who correctly identified the chatbot as 'capital' were unable to provide a precise explanation. They merely mentioned that the chatbot was 'man-made' without stating that it was 'used in production'. It should be noted that man-made products could be consumer goods, which are not usable as factors of production.
2(b)	Poor. The decision of the company to introduce chatbot (capital) to replace workers (labour) in answering commonly asked questions from customers was most relevant to the problem of 'how to produce', i.e., production method (or the combination of factor inputs to produce output). Only about one-third of the candidates were able to correctly identify the related basic economic problem as 'how to produce'. Some candidates provided irrelevant answers such as 'tertiary production' which were not related to the three basic economic problems.
3(a)	Fair. Roughly half of the candidates merely pointed out that the number of machines was fixed but failed to state more clearly that it was fixed with respect to output.
3(b)	Fair. Approximately one-third of the candidates were unable to calculate the marginal products correctly. Quite a number of candidates wrongly performed subtraction directly with the given figures of average outputs when computing the marginal products of labour. Some candidates even misinterpreted the given figures of 'average outputs' as 'marginal products'.
4	Satisfactory. About half of the candidates were able to explain well with two relevant reasons. However, some candidates did not realise that the question required them to analyse from customers' perspective, making it different from similar questions in the past. Reasons such as 'difference in production cost' were thus not acceptable.
5(a)	Fair. Quite a number of candidates did not understand that an increase in the marginal cost of producing each unit of a good implies an upward shift of the marginal cost (MC) curve. Some candidates were unable to indicate correctly in the diagram the specific area of decrease in total social surplus following the shift of the MC curve.
5(b)	Poor. A number of candidates agreed with the student's response, viz., the misconception that 'a fall in total social surplus implies inefficiency'. Some of them misinterpreted the fall in total social surplus in part (a) as a deadweight loss. It should be noted that inefficiency and deadweight loss would occur only if the final equilibrium somehow involves a discrepancy between marginal (social) cost and marginal (social) benefit.

Question Number	Performance in General
6	Satisfactory. Although most candidates were able to give the correct formula for computing the unemployment rate, quite a number of them did not seem to understand the relation among the three variables in the formula. For example, some candidates misunderstood that the unemployment rate would fall whenever the size of the labour force decreased. Some candidates failed to compare the percentage change in the labour force with that in the unemployed population.
7(a)	Satisfactory. Most candidates indicated the deflationary (output) gap accurately in the graph. However, some candidates mis-represented the output gap as an area (like deadweight loss). Some others did not even insert a long-run aggregate supply curve when attempting to show the output gap.
7(b)	Satisfactory. Most candidates correctly explained that the increase in the number of inbound tourists would lead to an increase in the exports of services. However, some candidates wrongly related the rise in the number of inbound tourists to an increase in private consumption expenditure.
8(a)	Satisfactory. Some candidates were careless in reading the question and did not realise that the amount of resources in Country A was twice of that in Country B.
8(b)	Fair. Quite a number of candidates mishandled the transportation cost. Some wrongly added the transportation cost to the lower-bound of the feasible range, effectively requiring the computer-exporting (rather than computer-importing) country to bear this cost. On the other hand, some candidates incorrectly added (rather than deducted) this cost to (from) the upper-bound of the range, hence expanding (rather than narrowing) the range of mutually beneficial terms of trade despite the fact that transportation cost was supposed to limit the gains from trade.
9(a)	Good.
9(b)	Fair. Some candidates were unable to distinguish between 'value' and 'volume' of imports. Moreover, a number of candidates failed to state clearly which currency (HKD or JPY) they were referring to in their answers.

Section B

Question Number	Performance in General
10(a)	Fair. Nearly half of the candidates were able to make use of concepts such as real interest rate and liquidity to answer the question. However, quite a number of candidates were unable to provide further elaborations. For example, some candidates simply stated that bonds would provide 'lower liquidity' than cash without any explanation.
10(b)(i)	Poor. Many candidates failed to comprehend the given information about the Silver Bonds and did not realise that its 4% nominal interest was a minimum return. When the expected inflation rate was above 4%, the expected nominal interest rate of the Silver Bonds would rise above 4% as well.
10(b)(ii)	Poor. Candidates who could not handle part (b)(i) naturally also performed poorly in part (b)(ii). Some candidates failed to explain the gain or loss of the bondholder by comparing the actual real rate of return with the expected real rate of return.
10(c)(i)	Fair. Quite a number of candidates were not familiar with the process of deposit contraction, and were thus unable to describe the process step-by-step.
10(c)(ii)	Poor. Only about half of the candidates provided a relevant reason relating to either excess reserves in banks or reduction in cash-holding by the public. Among them, only few candidates provided sufficient elaborations.
10(d)	Good.
10(e)	Fair. Most candidates mentioned the problem of using piece-rate contracts. But quite a number of them failed to relate these problems to the job nature of the Financial Secretary (e.g. outputs are not standardised / hard to be quantified).
11(a)	Satisfactory.
11(b)	Satisfactory. Most candidates were able to identify and explain the function of price illustrated in the given case. However, quite a number of candidates confused the 'functions of money' with the 'functions of price' and provided irrelevant answers.
11(c)	Fair. Many candidates correctly pointed out that the opportunity cost for the housewife to buy a ticket at the official ticketing office was lower than to buy it from touts, but did not make use of concepts such as time cost and money cost. Many candidates who did mention time cost and money cost failed nonetheless to compare the differences in terms of time cost and money cost of buying the ticket from touts vis-à-vis the ticketing office.
11(d)	Fair. Quite a number of candidates did not realise that the question was related to economies of scale. Some even showed confusion about 'revenue' and 'cost', and wrongly stated that an increase in revenue would lead to a decrease in (average) cost.
11(e)	Poor. Most candidates failed to explain the phenomenon with the concept of relative price. The question required an analysis of the change in relative price between higher-priced and lower-priced tickets when a lump-sum travelling expense incurred by fans from overseas was added to both kinds of tickets. Some candidates wrongly stated that incorporation of such lump-sum expense would lower the relative price of higher-priced concert tickets below that of lower-priced tickets.
12(a)(i)	Satisfactory. Some candidates were unable to make use of Source B to identify the relation between crude oil and gasoline and incorrectly considered crude oil and gasoline as in competitive supply or joint supply.
12(a)(ii)	Satisfactory. Most candidates correctly pointed out that the demand for gasoline was inelastic. However, some of them failed to apply the hints in Source B to account for the inelastic demand for gasoline.

Question Number	Performance in General
12(b)	Satisfactory. Most candidates were able to explain the relation between changes in money supply and changes in the general price level based on the quantity theory of money. However, some candidates showed difficulty in comparing the percentage change in money supply and that in real output while attempting to conclude about the change in the price level.
12(c)	Fair. Most candidates were able to relate the increase in discount rate to a fall in aggregate demand although many of them failed to explain explicitly how this effect came about via a reduction in money supply following the rise in the discount rate. Moreover, some candidates misunderstood the discount rate as the lending rate charged by commercial banks on loans made to the public. Some candidates even mistakenly interpreted the increase in discount rate as a fiscal policy.
12(d)	Fair. Most candidates presented their answers systematically and discussed the effects of the two strategies on each aspect of the economy in separate paragraphs. Some candidates were even able to analyse the effects from multiple perspectives. <u>Price level:</u> Most candidates demonstrated their ability to apply the AS-AD model to explain how each of the two strategies would affect the general price level. <u>Equity:</u> For Strategy 1, quite a number of candidates failed to point out that the low-income group was not affected by the strategy, and were thus unable to provide a comprehensive analysis. For Strategy 2, many candidates failed to compare the effect of the cancellation of tariffs in relation to the proportion of income of the lower-income group vis-à-vis that of the higher-income group. <u>Trade balance:</u> Some candidates did not explain why the value of imports would decrease under each of the strategies. A few concluded that there would be a trade surplus / deficit and did not realise that the original trade balance was not given.

General comments and recommendations

1. Some candidates lacked the acuity to apply economic concepts and models to analyse hypothetical or real-life problems. For example, quite a number of candidates failed to interpret the economic meaning of important hints in data sources. Candidates should strengthen their analytical skills instead of simply regurgitating concepts and theories.
2. Some candidates did not read the questions carefully. Candidates should devote special attention to the assumptions and conditions given in the questions to avoid giving irrelevant answers.
3. Some candidates seemed to have relied too heavily on memorisation of 'model answers' for past exam questions. They appeared to be reflexively recalling these 'model answers' when encountering similar questions without carefully interpreting the questions at hand, providing answers which were irrelevant.
4. Some candidates lacked awareness of current economic issues in our society. Candidates should pay more attention to economic news.
5. Some candidates lacked skills to extract information from data sources. Candidates should pay special attention to the headings, labels, and axes as they are essential to our understanding of the data presented therein.

Acknowledgements

The following material has been used in question papers in this volume:

Willis Towers Watson	Market Data: MPF market share as at March 31, 2021 https://www.wtwco.com/en-hk/insights/2021/04/mpf-market-share-as-at-march-31-2021
Census and Statistics Department	Consumer Price Indices https://www.censtatd.gov.hk/en/web_table.html?id=52
New Zealand Reseller News	NZ Post hires first digital employee courtesy of AWS partner Ambit https://www.reseller.co.nz/article/703314/nz-post-hires-first-digital-employee-courtesy-aws-partner-ambit/
Census and Statistics Department	Statistics on labour force, employment, unemployment and underemployment https://www.censtatd.gov.hk/en/web_table.html?id=210-06101#
Google Finance	United States Dollar to Japanese Yen https://www.google.com/finance/quote/USD-JPY?sa=X&ved=2ahUKEwj_rNmzipz_AhVV4WEKHZrTAPsQmY0JegQICxAc&window=1Y
e-Stat	Trade Statistics Data for Japan https://www.e-stat.go.jp/en/stat-search/files?page=1&layout=datalist&toukei=00350300&tstat=000001013141&cycle=1&year=20210&month=24101212&tclass1=000001013189&tclass2=000001013190&resultback=1&tclass3val=0
The Federal Reserve	Primary and Secondary credit https://www.frbdiscountwindow.org/Pages/Discount-Rates/Historical-Discount-Rates
The Federal Reserve	Money Stock Measures - H.6 Release https://www.federalreserve.gov/releases/h6/default.htm
Macrotrends	wti-crude-oil-prices-10-year-daily-chart www.macrotrends.net

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